ite your Enrollment Number) Enrollment No. 0521016201

End-Term Examination
(CBCS)(SUBJECTIVE TYPE)(OffLine)
Course Name:<B. Tech (MAE/DMAM)>, Semester:<3>
(Nov-Dec, 2022)

bject Code: BMA-201 Subject: Production Technology -I
ime : 3 Hours Maximum Marks :60
Note: Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.

(2.5*8=20)

Q1

(a) Casting is an important and prevalent form of Manufacturing process till date? Justify the statement?
(b) Explain the importance of riser in a gating system?
(c) What are the different flames in gas welding and how can by only seeing the three flames be known in Gas welding
(d) Discuss about porosity, a welding defect and its prevention?
(e) A manufacturing unit has to produce final part by reducing the diameter of the work piece and thus increasing the length. Which forming method is this?
(f) What are the advantages of powder metallurgy for manufacturing?
(g) Enumerate the difference between line standards and end standards?
(h) Discuss the application of the snap gauge?

UNIT-I

Q2 In casting, what is the usage of Cores. What is gating system and why it is used in casting? Explain in detail with a neat sketch. (10)

Q3 Centrifugal casting and Investment casting are special types of casting. Discuss both the processes in detail with neat sketches. (10)

UNIT-II

Q4 Elaborate the principle of the oxyacetylene welding with a neat sketch. Also discuss the role of mixing ratios in case of this gas welding. (10)

Q5 What is the role of the welding positions in achieving different joints. Explain the principle of Electron Beam Welding along with its diagram and applications? (10)

UNIT-III

Q6 Discuss how hot forming is different from cold forming? Also elaborate the three types of forging processes along with their applications and advantages? (10)

Q7 3D Printing and Additive manufacturing go hand in hand. Elaborate what is the difference 3D Printers and Additive Manufacturing? Explain the working of 3D printing. (10)

UNIT-IV

Q8 Explain how length standards and end standards are different methods of measurement. Discuss how length standards can be transferred to end standards. (10)

Q9 Slip gauges are very commonly used in various measurement labs. Illustrate the applications of slip gauges in measurement labs. Also discuss about Limit gauge. (10)

Q10) What is powder Metallurgy process explain along with a diagram? Illustrate the various steps involved in formation of powder in Powder Metallurgy. Explain why atomization and sintering is needed in Powder Metallurgy
(10 marks)

Smith

powder